ABES Engineering College.
Solution of Fundamentals of Linear Algebra & Statistics.
Mid-Term-1, Session: 2025-26
[25AS102]

1(a) Sum of Eigenvalues

= Trace of Matrix

= Sum of diagonal elements

$= 1 + 2 + 3 = 6$ —(0.5)

Product of Eigenvalues

= determinant of Matrix

$= 1(6-0) - 6(3-0) + 1(0-0)$

$= 6 - 18 = -12$ —(0.5)

1(b) Every square matrix satisfies its own characteristic equation. —(1)

1(c) $A = \begin{bmatrix} 2 & 1 & 2 \\ 1 & 1 & 3 \\ 4 & 3 & b \end{bmatrix}$; $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

$b = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

System has non-trivial solution if rank$(A) \neq 3$ —(0.5)

Echelon form of A:

$$\begin{bmatrix} 2 & 1 & 2 \\ 1 & 1 & 3 \\ 4 & 3 & b \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 3 \\ 2 & 1 & 2 \\ 4 & 3 & b \end{bmatrix} R_1 \leftrightarrow R_2$$

$$\sim \begin{bmatrix} 1 & 1 & 3 \\ 0 & -1 & -4 \\ 0 & -1 & b-12 \end{bmatrix} \begin{matrix} R_2 \to R_2 - 2R_1 \\ R_4 \to R_4 - 4R_1 \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 1 & 3 \\ 0 & -1 & -4 \\ 0 & 0 & b-8 \end{bmatrix} R_3 \to R_3 - R_2$$

$\Rightarrow$ The system has non-trivial solution if $b - 8 = 0$

$\Rightarrow b = 8$ —(0.5)

1(d) Consider the polynomial $p_1 = 1 + x^2$ & $p_2 = 1 - x^2$

then $p_1, p_2 \in P_2(x)$

where $P_2(x)$ is the collection of all the polynomials of degree 2.

But $p_1 + p_2 = 2 \notin P_2(x)$

$\Rightarrow P_2(x)$ is not closure for vector addition

$\Rightarrow P_2(x)$ is not a vector space. —(1)

1(e) Let V be a vector space over a field F. A non-empty subset W of V is called a subspace of V if W is also a vector space over F for the same vector addition & scalar multiplication of V over F.

Example: for the vector space $\mathbb{R}^3$ over $\mathbb{R}$. —(1)

$W_1 = \{(a, b, 0) : a, b \in \mathbb{R}\}$ is a subspace of $\mathbb{R}^3$.

$W_2 = \{(a, 0, a^2) : a \in \mathbb{R}\}$ is not a subspace of $\mathbb{R}^3$.

Q12. (a). $A = \begin{bmatrix} 1 & 2 & -1 & 4 \\ 2 & 4 & 3 & 4 \\ 1 & 2 & 3 & 4 \\ -1 & -2 & 6 & -7 \end{bmatrix}$

$\rho(A)$ = order of Identity matrix in canonical form. (0.5)

②

$R_2 \leftrightarrow R_2 - 2R_1$, $R_3 \to R_3 - R_1$, $R_4 \to R_4 + R_1$

$$\sim \begin{bmatrix} 1 & 2 & -1 & 4 \\ 0 & 0 & 5 & -4 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 5 & -3 \end{bmatrix}$$

$C_2 \to C_2 - 2C_1$, $C_3 \to C_3 + C_1$, $C_4 \to C_4 - 4C_1$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 5 & -4 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 5 & -3 \end{bmatrix} \Rightarrow C_2 \leftrightarrow C_4$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -4 & 5 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & -3 & 5 & 0 \end{bmatrix} \Rightarrow C_3 \to C_3 + C_2$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -4 & 1 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & -3 & 2 & 0 \end{bmatrix} \Rightarrow R_4 \to R_4 - 2R_2$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -4 & 1 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 5 & 0 & 0 \end{bmatrix} \Rightarrow R_4 \to R_4 + R_2, R_3 \to \frac{1}{4} R_3$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -4 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix} \Rightarrow R_2 \leftrightarrow R_4$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -4 & 1 & 0 \end{bmatrix} \Rightarrow R_4 \to R_4 + 4R_2 \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 5 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 5 & 0 \end{bmatrix} \Rightarrow R_4 \to R_4 - 5R_3$$

$$\sim \left[\begin{array}{ccc:c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \hdashline 0 & 0 & 0 & 0 \end{array}\right] = \begin{bmatrix} I_3 & : & 0_{3\times1} \\ 0_{1\times3} & : & 0_{1\times1} \end{bmatrix} \text{— (1)}, \quad r(A) = 3 \text{ — (0.5)}$$

58
21
79

31
17
48
21
69

Sol2 (b)

$$A = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 2 & 1 \\ 5 & 2 & -3 \end{bmatrix}$$

$A = IA$ (0.5)

$$\begin{bmatrix} 2 & 1 & -1 \\ 0 & 2 & 1 \\ 5 & 2 & -3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

$R_3 \to R_3 - 2R_1$

$$\sim \begin{bmatrix} 2 & 1 & -1 \\ 0 & 2 & 1 \\ 1 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} A$$

$R_3 \leftrightarrow R_1$

$$\sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 2 & 1 \\ 2 & 1 & -1 \end{bmatrix} = \begin{bmatrix} -2 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} A$$

$R_3 \to R_3 - 2R_1$

$$\sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 0 & 1 \\ 0 & 1 & 0 \\ 5 & 0 & -2 \end{bmatrix} A$$

$R_2 \leftrightarrow R_3$

$$\sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 2 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 0 & 1 \\ 5 & 0 & -2 \\ 0 & 1 & 0 \end{bmatrix} A$$

$R_3 \to R_3 - 2R_2$

$$\sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} -2 & 0 & 1 \\ 5 & 0 & -2 \\ -10 & 1 & 4 \end{bmatrix} A$$

$R_1 \to R_1 - R_3, \ R_2 \to R_2 + R_3$

$$\sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 8 & -1 & -3 \\ -5 & 1 & 2 \\ -10 & 1 & 4 \end{bmatrix} A$$

$R_3 \to -R_3$

$$\sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 8 & -1 & -3 \\ -5 & 1 & 2 \\ 10 & -1 & -4 \end{bmatrix} A$$

$$A^{-1} = \begin{bmatrix} 8 & -1 & -3 \\ -5 & 1 & 2 \\ 10 & -1 & -4 \end{bmatrix}$$

— (1.5)

Linear equations

$x+2y+3z=6$
$x+3y+5z=9$
$2x+5y+az=b$, have

1) no soln.
2) unique soln
3) Infinitely many soln.

$AX=B$. $\begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 5 \\ 2 & 5 & a \end{bmatrix} = \begin{bmatrix} 6 \\ 9 \\ b \end{bmatrix}$

consider Augumented matrix

$A:B = \begin{bmatrix} 1 & 2 & 3 & : & 6 \\ 1 & 3 & 5 & : & 9 \\ 2 & 5 & a & : & b \end{bmatrix}$

for rank.

$\xrightarrow[R_3-2R_1]{R_2-R_1} \begin{bmatrix} 1 & 2 & 3 & : & 6 \\ 0 & 1 & 2 & : & 3 \\ 0 & 1 & a-6 & : & b-12 \end{bmatrix}$

$\xrightarrow{R_3-R_2} \begin{bmatrix} 1 & 2 & 3 & : & 6 \\ 0 & 1 & 2 & : & 3 \\ 0 & 0 & a-8 & : & b-15 \end{bmatrix}$ —— (0.5)

1) For No soln

$\rho(A) \neq \rho(A:B)$

i.e. if $a-8=0$, $b-15 \neq 0$

$\boxed{a=8, \ b \neq 15}$ —— (0.5)

2) For Unique soln

$\rho(A) = \rho(A:B) = $ no. of variables $=3$

i.e. $a-8 \neq 0$, b can take any value.

[$a \neq 8$, b can take any value] —— (0.5)

3) For infinite soln. $\rho(A) = \rho(A:B) \neq$ no of variables

i.e. $\boxed{a=8}$ $\boxed{b=15}$ —— (0.5)

Section-C

$$A = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 2 & 1 \\ -1 & 2 & 2 \end{bmatrix}$$

characteristic eqn $|A - \lambda I| = 0$

$$\begin{vmatrix} 1-\lambda & 2 & 2 \\ 0 & 2-\lambda & 1 \\ -1 & 2 & 2-\lambda \end{vmatrix} = 0$$

$$\lambda^3 - (\text{trace } A)\lambda^2 + [C_{11}+C_{22}+C_{33}]\lambda - |A| = 0$$

$$\lambda^3 - 5\lambda^2 + 8\lambda - 4 = 0$$

$$\boxed{\lambda = 1, 2, 2} \text{ —— (1)}$$

eigen vector $[A - \lambda I]X = 0$

for $\lambda = 1$ $$\begin{bmatrix} 0 & 2 & 2 \\ 0 & 1 & 1 \\ -1 & 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$0x + 2y + 2z = 0$

$-x + 2y + z = 0$

$$\frac{x}{2-4} = \frac{-y}{0+2} = \frac{z}{0+2}$$

$$X_1 = \begin{bmatrix} -2 \\ -2 \\ 2 \end{bmatrix} = k\begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \text{ —— (1)}$$

for $\lambda = 2$ $$\begin{bmatrix} -1 & 2 & 2 \\ 0 & 0 & 1 \\ -1 & 2 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$-x + 2y + 2z = 0$

$z = 0$

for $\lambda = 2$ $$X_2 = k\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \text{ —— (1)}$$

Q.3 b) $$A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$$

soln characteristic eqn $|A - \lambda I| = 0$

$$\lambda^3 - 5\lambda^2 + 7\lambda - 3I = 0 \text{ — (1)}$$

To verify cayley Hamilton theorem we have to show A satisfies its own characteristic eqn.

ie $A^3 - 5A^2 + 7A - 3I = 0$

$$A^2 = \begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix} \quad A^3 = \begin{bmatrix} 14 & 13 & 13 \\ 0 & 1 & 0 \\ 13 & 13 & 14 \end{bmatrix}$$

$$\boxed{A^3 - 5A^2 + 7A - 3I = 0} \text{ — (1)}$$

—— (1)

To find A^{-1}, from eqn (1)

$$A^3 - 5A^2 + 7A - 3I = 0$$

$$I = \frac{1}{3}[A^3 - 5A^2 + 7A]$$

Multiply by A^{-1}

$$A^{-1} = \frac{1}{3}[A^2 - 5A + 7I]$$

$$A^{-1} = \frac{1}{3}\begin{bmatrix} 2 & -1 & -1 \\ 0 & 3 & 0 \\ -1 & -1 & 2 \end{bmatrix} \text{ —— (1)}$$

$y = -z$

$-x + 2y + z = 0$

$x + z = 0$

$x = -z$

$$\begin{bmatrix} -z \\ -z \\ z \end{bmatrix}$$

1

1

-1

Section-C

Let $V = R^3$ be a vector space and $W = \{(x,y,z) : x-3y+4z=0\}$. Prove that W is a subspace of V over R.

Sol^n W is a subspace of V over R if

1) $0 \in W$

$0 = (0,0,0)$, $0 \in W$ if

$0 - 3\times 0 + 4 \times 0 = 0$

LHS = RHS $\Rightarrow$ $0 \in W$ —(1)

2) $au+bv \in W$; $\forall a,b \in R$, $u,v \in W$

Let $u \in W \Rightarrow u = (x_1,y_1,z_1)$: $x_1 - 3y_1 + 4z_1 = 0$

$v \in W \Rightarrow v = (x_2, y_2, z_2)$; $x_2 - 3y_2 + 4z_2 = 0$

consider

$au+bv$

$a(x_1,y_1,z_1) + b(x_2,y_2,z_2)$

$au+bv \in W$

if $ax_1 + bx_2 - 3(ay_1+by_2) + 4(az_1+bz_2) = 0$

LHS $a(x_1-3y_1+4z_1) + b(x_2-3y_2+4z_2)$

$\Rightarrow a(0) + b(0)$

$= 0 =$ RHS.

$\Rightarrow au+bv \in W$ —(2)

Hence W is a subspace of R^3 over R.

Q4b) Prove that set of all vectors in a plane (R^2) over the field of real numbers is a vector space with respect to vector addition and scalar multiplication.

Sol. Let $V = \{(x,y); x,y \in R\}$.

To prove V is a vector space over R under vector addition operation and scalar multiplication if it satisfies following properties.

1. Closure. $\forall u,v \in V$

$u+v \in V$

Let $u = (x_1,y_1)$; $x_1,y_1 \in R$

$v = (x_2,y_2)$; $x_2,y_2 \in R$

$u+v = (x_1+x_2, y_1+y_2) \in V$ $\forall x_1,x_2,y_1,y_2 \in R$

2. Commutative

$u+v = v+u$

$u+v = (x_1+x_2, y_1+y_2)$

$= (x_2+x_1, y_2+y_1)$

$= (x_2,y_2) + (x_1,y_1)$

$= v+u$

$\Rightarrow$ V is commutative with respect to addition

3. Associative $\forall u,v,w \in V$.

$u+(v+w) = (u+v)+w$

LHS $(x_1,y_1) + ((x_2,y_2)+(x_3,y_3))$

$\Rightarrow (x_1+x_2+x_3, y_1+y_2+y_3)$

RHS $(u+v)+w = ((x_1,y_1)+(x_2,y_2)) + (x_3,y_3)$

$= (x_1+x_2+x_3, y_1+y_2+y_3)$

LHS = RHS.

V is associative w.r.to. addition

7

...itive Identity

$u \in V$. there exists $0 \in V$ such that $u+0 = 0+u = u$

let $0 = (0,0)$; $0 \in R$

$u+0 = (x_1, y_1) + (0,0)$

$= (x_1+0, y_1+0) = (x_1, y_1)$

$= u = RHS$.

5. Additive Inverse.

$\forall u \in V$ there exists $v \in V$ such that $u+v = 0$, $0 \in V$.

$(x_1, y_1) + (x_2, y_2) = (0,0)$

$\Rightarrow (x_2, y_2) = (0,0) - (x_1, y_1)$

$\Rightarrow \forall (x_1, y_1) \exists$

$x_2 = -x_1$, $y_2 = -y_1$

such that $u+v = 0$ — (1.5)

6. Closure w.r.to multiplication.

$\forall u, \in V$, $a \in R$.

$a \cdot u \in V$.

$a \cdot (x_1, y_1) = (ax_1, ay_1)$

$\in V$

$\Rightarrow$ V is closed under scalar multiplication.

7. $\forall u, v \in V, a \in R$

$a \cdot (u+v) = a.u + a.v$

LHS $a \cdot (x_1+x_2, y_1+y_2)$

$= (ax_1 + ax_2, ay_1 + ay_2)$

$= (ax_1, ay_1) + (ax_2, ay_2)$

$= a.u + a.v$

8. $\forall a, b \in R$, $u \in V$

$(a+b) \cdot u = a.u + b.u$

LHS $(a+b) \cdot (x_1, y_1) = ((a+b)x_1, (a+b)y_1)$

$= (ax_1, ay_1) + (bx_1, by_1)$

$= a \cdot u + b \cdot u = RHS$.

9. For $u \in V$ and, $a, b \in R$

$a(bu) = (ab)u$

LHS $a(b(x_1, y_1))$

$= a(bx_1, by_1)$

$= (abx_1, aby_1)$

$= (ab)u = RHS$.

10. Multiplicative identity.

$\forall u \in V$ there exists $1 \in R$ such that

$1 \cdot u = u \cdot 1 = u$

LHS.

$1 \cdot (x_1, y_1) = (x_1, y_1) = u$

Hence all the properties are satisfied — (1.5)

$\therefore V = R^2$ is a vector space with respect to vector addition and scalar multiplication over the field of real nos. R.